

ASSIGNMENT #03

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CHE-301 Assignment-03

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1 Abstract

This report entails a thermodynamic analysis of nitrogen and ethylene systems under both non-ideal and ideal gas conditions. In Problem A, Compressibility Factor data for nitrogen was used to evaluate thermodynamic properties at 10°C and 830 bar. The thermal expansion coefficient, pressure expansion coefficient, molar enthalpy and entropy, heat capacities under constant pressure and volume, Joule–Thomson coefficient, and speed of sound were determined.

In Problem B, the unsteady venting of ethylene from a pressurized tank was analyzed assuming ideal gas behavior. Mass, energy, and entropy balances were used to determine the initial mass, critical flow conditions, nozzle diameter, and remaining mass in the tank at a varied pressure.

2 Data Summary

Question A - Thermodynamic Properties of Nitrogen

Variable	Property	Value	Units
β	Thermal Expansion Coefficient	0.001888	K^{-1}
ζ	Pressure Expansion Coefficient	0.005201	K^{-1}
H	Molar Enthalpy	1.5271	$\frac{kJ}{mol}$
S	Molar Entropy	-9.4703	$\frac{kJ}{mol \cdot K}$
C_p	Molar Heat Capacity (Constant Pressure)	34.7120	$\frac{kJ}{mol \cdot K}$
C_v	Molar Heat Capacity (Constant Volume)	22.97023	$\frac{kJ}{mol \cdot K}$
μ_{JT}	Joule-Thomson Coefficient	-68.19730	$\frac{K}{bar}$
ν	Speed of Sound	792.0596	$\frac{m}{s}$

Question B - Venting of Ethylene Gas

Variable	Property	Value	Units
m_0	Initial Mass of Ethylene	3,374.1206	kg
P_s	Pressure at Critical Condition	113.6321	bar
T_s	Temperature at Critical Condition	93.6246	$^{\circ}C$
A	Narrowed Pipe Area	37.0790	mm^2
d	Narrowed Pipe Diameter	6.8710	mm
m_c	Mass at Critical Condition	2,090.7038	kg
m_p	Mass at Pressure P	2,787.2552	kg

3 Question A

For this question, we are asked to solve for the following at a temperature of 10°C and a pressure of 830 bar

1. The thermal expansion coefficient in K^{-1}
2. The pressure expansion coefficient in K^{-1}
3. The molar enthalpy in $\frac{\text{kJ}}{\text{mol}}$
4. The molar entropy in $\frac{\text{kJ}}{\text{kmol}\cdot\text{K}}$
5. The molar heat capacity CP in $\frac{\text{kJ}}{\text{kmol}\cdot\text{K}}$
6. The molar heat capacity CV in $\frac{\text{kJ}}{\text{kmol}\cdot\text{K}}$
7. The Joule-Thomson coefficient in $\frac{\text{K}}{\text{Bar}}$
8. The speed of sound in $\frac{\text{m}}{\text{s}}$

The initial step to solving for these values was to import the given data into a CHE Calculator excel file. The data has been listed below

P (atm)	Temperature (deg C)				
	0	25	50	75	100
0	1.000	1.000	1.000	1.000	1.000
1	1.000	1.000	1.000	1.000	1.000
3	0.999	0.999	1.000	1.000	1.001
5	0.998	0.999	1.000	1.001	1.001
10	0.996	0.998	1.000	1.001	1.002
30	0.959	0.996	1.001	1.005	1.007
50	0.958	0.996	1.004	1.009	1.014
100	0.984	1.004	1.018	1.027	1.033
200	1.036	1.057	1.072	1.082	1.089
300	1.132	1.146	1.154	1.160	1.163
400	1.256	1.254	1.253	1.250	1.247
600	1.524	1.495	1.471	1.450	1.432
800	1.798	1.723	1.697	1.658	1.625
1000	2.067	1.988	1.923	1.867	1.819

Figure 1: Initial Table values

In order to continue with the question, I interpolated the values to determine the $Z = \text{compressibility factor}$ for both the column for 10°C and the row for 830 bar. There is an initial step that I needed to do before the Interpolation, and that was to convert the units for pressure. The data given in the chart is in terms of atm and the given pressure is in bar.

$$P(\text{bar}) = 830 \quad \therefore \quad P(\text{atm}) = 819.15$$

I was able to convert this value directly in excel through the CHE calculator's command "=BAR-toATM(input)" where input is the cell containing the pressure in bar.

Using the equation below, I can interpolate the Z .

$$y = y_1 + \frac{x - x_1}{x_2 - x_1} * (y_2 - y_1)$$

Applying the equation in terms of this questions yields,

$$Z = Z_1 + \frac{T - T_1}{T_2 - T_1} * (Z_2 - Z_1) \quad (1)$$

Pressure (atm)	Compressibility Factor					
	Temperature					
	0	10	25	50	75	100
0	273.15	283.15	298.15	323.15	348.15	373.15
0	1.00000		1.00000	1.00000	1.00000	1.00000
1	1.00000		1.00000	1.00000	1.00000	1.00000
3	0.99900		0.99900	1.00000	1.00000	1.00100
5	0.99800		0.99900	1.00000	1.00100	1.00100
10	0.99600		0.99800	1.00000	1.00100	1.00200
30	0.95900		0.99600	1.00100	1.00500	1.00700
50	0.95800		0.99600	1.00400	1.00900	1.01400
100	0.98400		1.00400	1.01800	1.02700	1.03300
200	1.03600		1.05700	1.07200	1.08200	1.08900
300	1.13200		1.14600	1.15400	1.16000	1.16300
400	1.25600		1.25400	1.25300	1.25000	1.24700
600	1.52400		1.49500	1.47100	1.45000	1.43200
800	1.79800		1.72300	1.69700	1.65800	1.62500
819.15						
1000	2.06700		1.98800	1.92300	1.86700	1.81900

Figure 2: Parted data table before interpolation

I applied Equation One in the above chart down the column for 10°C , which gives the Z values corresponding to this constant temperature and varying pressures. This defines the equation variables as

Z = compressibility factor for 10°C

Z_1 = compressibility factor for 0°C

Z_2 = compressibility factor for 25°C

T = compressibility factor for 10°C

T_1 = compressibility factor for 0°C

T_2 = compressibility factor for 25°C

Next, I needed to interpolate the Z values across varying temperatures with a constant pressure of 819.15 bar. In this case the equation will be rewritten below as

$$Z = Z_1 + \frac{P - P_1}{P_2 - P_1} * (Z_2 - Z_1) \quad (2)$$

Z = compressibility factor for 819.15 atm

Z_1 = compressibility factor for 800 atm

Z_2 = compressibility factor for 1000 atm

P = compressibility factor for 819.15 atm

P_1 = compressibility factor for 800 atm

P_2 = compressibility factor for 1000 atm

Plugging these formulas in the CHE Calculator excel table gives the following tabulated data for this question, where the highlighted cell represents the Z value associated with 10°C and 819.15 atm

Pressure (atm)	Compressibility Factor					
	Temperature					
	0	10	25	50	75	100
0	273.15	283.15	298.15	323.15	348.15	373.15
0	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000
1	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000
3	0.99900	0.99900	0.99900	1.00000	1.00000	1.00100
5	0.99800	0.99840	0.99900	1.00000	1.00100	1.00100
10	0.99600	0.99680	0.99800	1.00000	1.00100	1.00200
30	0.95900	0.97380	0.99600	1.00100	1.00500	1.00700
50	0.95800	0.97320	0.99600	1.00400	1.00900	1.01400
100	0.98400	0.99200	1.00400	1.01800	1.02700	1.03300
200	1.03600	1.04440	1.05700	1.07200	1.08200	1.08900
300	1.13200	1.13760	1.14600	1.15400	1.16000	1.16300
400	1.25600	1.25520	1.25400	1.25300	1.25000	1.24700
600	1.52400	1.51240	1.49500	1.47100	1.45000	1.43200
800	1.79800	1.76800	1.72300	1.69700	1.65800	1.62500
819.15	1.82375	1.79360	1.74837	1.71864	1.67801	1.64357
1000	2.06700	2.03540	1.98800	1.92300	1.86700	1.81900

Figure 3: Interpolated Compression Factor Table

The next step is to transform the data table above to be in terms of ρ = density instead of Z = compressibility factor. I started with the definition of Z , where

$$Z = \frac{P}{\rho RT}$$

P = Pressure in atm

ρ = density in $\frac{kmol}{m^3}$

T = Temperature in k

$$R = \text{Gas Constant} = 0.082057 \frac{L * atm}{mol * K}$$

Rearranging, we find the expression

$$\rho = \frac{P}{ZRT}$$

It is important to note that the given pressure of 830 bar was converted to atm, but the temperature was kept as celcius. In the following transform, the temperature will be converted to kelvin, this is done by the calculation below. This is in an effort to simplify further calculations and to not require future unit conversions.

$$K(kelvin) = T^{\circ}C + 273.15$$

We can apply this equation to a new table, with the same axis of pressure and temperature. For an example,

$$\rho_1 = \frac{P_1}{Z_1 RT_1}$$

where ρ_1 is corresponding to the same temperature and pressure as the Z value is associated with. This gives the following chart of tabulated values. The units of density in this chart is in terms of $\frac{mol}{L}$, this will be converted to $\frac{kmol}{m^3}$.

$$1 \frac{mol}{L} * \frac{1000L}{1m^3} * \frac{1kmol}{1000mol} = 1 \frac{kmol}{m^3}$$

Therefore, there is no need for a unit conversion in the calculations as the ratio is 1:1 conversion. So, we can treat the table output as density in terms of $\frac{kmol}{m^3}$

	Density					
	Temperature					
	0	10	25	50	75	100
Pressure (atm)	273.15	283.15	298.15	323.15	348.15	373.15
0	0.000	0.000	0.000	0.000	0.000	0.000
1	0.045	0.043	0.041	0.038	0.035	0.033
3	0.134	0.129	0.123	0.113	0.105	0.098
5	0.224	0.216	0.205	0.189	0.175	0.163
10	0.448	0.432	0.410	0.377	0.350	0.326
30	1.396	1.326	1.231	1.130	1.045	0.973
50	2.329	2.211	2.052	1.878	1.735	1.610
100	4.534	4.339	4.071	3.705	3.408	3.162
200	8.613	8.242	7.734	7.036	6.470	5.998
300	11.824	11.350	10.700	9.804	9.053	8.424
400	14.209	13.716	13.038	12.039	11.201	10.476
600	17.565	17.075	16.404	15.382	14.484	13.684
800	19.851	19.475	18.978	17.778	16.890	16.078
819.15	20.039	19.656	19.150	17.975	17.088	16.277
1000	21.585	21.146	20.560	19.611	18.749	17.954

Figure 4: Density Table

3.1 β , Thermal Expansion Coefficient

Then, I moved on to solving for β = **Thermal Expansion Coefficient**. For this question, I defined β as

$$\beta = \frac{1}{V} \left(\frac{\partial V}{\partial T} \right)_P = -\frac{1}{\rho} \left(\frac{\partial \rho}{\partial T} \right)_P \quad (3)$$

We can derive this relation starting with

$$\begin{aligned} \rho V &= 1 \\ \rho dV + V d\rho &= 0 \\ \rho dV &= -V d\rho \\ \frac{dV}{V} &= -\frac{d\rho}{\rho} \end{aligned}$$

Plugging this relation into the center equation above yields the equation to the right.

In this case, the ρ is the density for the selected cell for 819.15 atm and 10°C. In order to solve for the partial derivative, I used the "fAkima(XData,F(x)data, Job)" function in the CHE Calculator.

I set the range of temperatures as the xData and the range of density ρ as the F(x)Data, then set the Job = 3 as this solves for the first derivative. The values of ρ are kept at constant pressure of 819.15 atm. This returned,

$$\left(\frac{\partial \rho}{\partial T} \right)_P = -0.037117791$$

Plugging this into equation 3 along side the density returns a thermal coefficient of

$$\beta = 0.001888331 K^{-1}$$

3.2 ζ , Pressure Expansion Coefficient

In order to solve for the **Pressure Expansion Coefficient**, I started with the equation definition

$$\zeta = \frac{1}{P} \left(\frac{\partial P}{\partial T} \right)_\rho \quad (4)$$

In order to solve for this, I first had to solve for an alternate partial as I don't have a constant density for this partial. To do this, I returned back to the definition of β

$$\beta = -\frac{1}{\rho} \left(\frac{\partial \rho}{\partial T} \right)_P$$

I applied the cyclic and reciprocal identities for partials derivatives and now have

$$\left(\frac{\partial \rho}{\partial T} \right)_P \left(\frac{\partial T}{\partial P} \right)_\rho \left(\frac{\partial P}{\partial \rho} \right)_T = -1$$

Rearranging with the cyclic and reciprocity identities, I get the following equation

$$\left(\frac{\partial \rho}{\partial T} \right)_P = -\frac{\left(\frac{\partial P}{\partial T} \right)_\rho}{\left(\frac{\partial P}{\partial \rho} \right)_T}$$

I can now plug this in to equation 3 which yields

$$\beta = \frac{1}{\rho} \frac{\left(\frac{\partial P}{\partial T} \right)_\rho}{\left(\frac{\partial P}{\partial \rho} \right)_T}$$

I then rearranged this equation to solve for the partial over a constant temperature

$$\left(\frac{\partial P}{\partial T} \right)_\rho = \beta \rho \left(\frac{\partial P}{\partial \rho} \right)_T \quad (5)$$

I then returned to equation 4 and plug in the above relation and this yields a final equation for the **Pressure Expansion Coefficient** of

$$\zeta = \frac{\beta \rho}{P} \left(\frac{\partial P}{\partial \rho} \right)_T \quad (6)$$

I was now able to solve this similarly to the solution of the **Thermal Expansion Coefficient**

$$\beta = 0.001888331 K^{-1}$$

$$\rho_{819.15 \text{ atm}, 10^\circ C} = 19.656 \frac{\text{kmol}}{\text{m}^3}$$

$$P = 819.15 \text{ atm}$$

$$\left(\frac{\partial P}{\partial \rho} \right)_T = \text{"=fAkima(x,y,job)"} = 114.7742198$$

For the partial, the XData was set to be the varying density down the constant column of $10^\circ C$ and the F(x)Data was set to the varying pressure column. Plugging all these values into equation 6 yields

$$\zeta = 0.005200738 K^{-1}$$

3.3 C_p and C_v , Molar Heat Capacities

In class we derived an expression for the difference of $C_p - C_v$

$$C_p - C_v = \frac{T \left(\frac{\partial P}{\partial T} \right)_\rho^2}{\rho^2 \left(\frac{\partial P}{\partial \rho} \right)_T} \quad (7)$$

Here I ran into the same issue as solving for the **Pressure Expansion Coefficient**, however we already solved for an equivalent partial, recall equation 5 and plugging into equation 7 gives a new expression of

$$C_p - C_v = \frac{T \left(\beta \rho \left(\frac{\partial P}{\partial \rho} \right)_T \right)^2}{\rho^2 \left(\frac{\partial P}{\partial \rho} \right)_T}$$

This can be simplified as below

$$\begin{aligned} C_p - C_v &= \frac{T \beta^2 \rho^2 \left(\frac{\partial P}{\partial \rho} \right)_T^2}{\rho^2 \left(\frac{\partial P}{\partial \rho} \right)_T} \\ C_p - C_v &= T \beta^2 \left(\frac{\partial P}{\partial \rho} \right)_T \end{aligned} \quad (8)$$

Recalling equation 6, we can rearrange 6 and combine with 8 and get a simplified equation below

$$C_p - C_v = \frac{T \beta P \zeta}{\rho}$$

where we know the following,

$$T = 10^\circ C = 283.15 K$$

$$\beta = 0.001888331 K^{-1}$$

$$\zeta = 0.005200738 K^{-1}$$

$$\rho_{819.15 \text{ atm}, 10^\circ C} = 19.656 \frac{\text{kmol}}{\text{m}^3}$$

$$P = 819.15 \text{ atm} = 83,000 \text{ kPa}$$

Plugging all these values yields

$$C_p - C_v = 11.7417745 \frac{\text{kJ}}{\text{kmol} \cdot K}$$

In class we derived an equation for the Constant Volume heat Capacity, below

$$C_v = C_v^{IG} + \int_{\infty}^V T \left(\frac{\partial^2 P}{\partial T^2} \right)_\rho dV \quad (9)$$

3.3.1 Evaluation of $\left(\frac{\partial^2 P}{\partial T^2} \right)_\rho$

In order to solve for this partial $\left(\frac{\partial^2 P}{\partial T^2} \right)_\rho$, I started with its first derivative

$$\left(\frac{\partial P}{\partial T} \right)_\rho = - \frac{\left(\frac{\partial \rho}{\partial T} \right)_P}{\left(\frac{\partial \rho}{\partial P} \right)_T} = - \frac{\rho_T}{\rho_P}$$

Taking the second derivative and applying the quotient rule,

$$\begin{aligned} \left(\frac{\partial P}{\partial T} \right)_\rho &= - \frac{\rho_T}{\rho_P} \\ \left(\frac{\partial^2 P}{\partial T^2} \right)_\rho &= \frac{\rho_P * \left(\frac{\partial \rho_T}{\partial T} \right)_\rho - \rho_T * \left(\frac{\partial \rho_P}{\partial T} \right)_\rho}{\rho_P^2} \end{aligned} \quad (10)$$

Returning to equation 11, I plugged in all the solved for differentials and got

3.3.2 Solving for $\left(\frac{\partial \rho_T}{\partial T}\right)_\rho$

Since ρ_T and ρ_P still depend on both T and P , the derivatives at constant density are functions of P with T :

$$d(\rho_T) = \left(\frac{\partial \rho_T}{\partial T}\right)_P dT + \left(\frac{\partial \rho_T}{\partial P}\right)_T dP$$

Knowing that

$$\begin{aligned} \left(\frac{dP}{dT}\right)_\rho &= \left(\frac{\partial P}{\partial T}\right)_\rho \\ dP &= \left(\frac{\partial P}{\partial T}\right)_\rho dT \end{aligned}$$

Plugging this in above yields

$$\begin{aligned} d(\rho_T) &= \left(\frac{\partial \rho_T}{\partial T}\right)_P dT + \left(\frac{\partial \rho_T}{\partial P}\right)_T \left(\frac{\partial P}{\partial T}\right)_\rho dT \\ d(\rho_T) &= \left(\left(\frac{\partial \rho_T}{\partial T}\right)_P + \left(\frac{\partial \rho_T}{\partial P}\right)_T \left(\frac{\partial P}{\partial T}\right)_\rho \right) dT \\ \left(\frac{\partial \rho_T}{\partial T}\right)_\rho &= \left(\frac{\partial \rho_T}{\partial T}\right)_P + \left(\frac{\partial \rho_T}{\partial P}\right)_T \left(\frac{\partial P}{\partial T}\right)_\rho \end{aligned}$$

Now, subbing in the values for the differentials of ρ only in terms of ρ , not its partials,

$$\begin{aligned} \rho_T &= \left(\frac{\partial \rho}{\partial T}\right)_P \\ \left(\frac{\partial \rho_T}{\partial T}\right)_P &= \left(\frac{\partial}{\partial T}\right)_P * \left(\frac{\partial \rho}{\partial T}\right)_P = \left(\frac{\partial^2 \rho}{\partial T^2}\right) = \rho_{TT} \\ \left(\frac{\partial \rho_T}{\partial P}\right)_T &= \left(\frac{\partial}{\partial P}\right)_T * \left(\frac{\partial \rho}{\partial T}\right)_P = \left(\frac{\partial^2 \rho}{\partial T \partial P}\right) = \rho_{TP} \\ \therefore \left(\frac{\partial \rho_T}{\partial T}\right)_\rho &= \rho_{TT} + \rho_{TP} \left(\frac{\partial P}{\partial T}\right)_\rho = \end{aligned}$$

3.3.3 Solving for $\left(\frac{\partial \rho_P}{\partial T}\right)_\rho$

$$d(\rho_P) = \left(\frac{\partial \rho_P}{\partial T}\right)_P dT + \left(\frac{\partial \rho_P}{\partial P}\right)_T dP$$

Knowing that

$$\begin{aligned} \left(\frac{dP}{dT}\right)_\rho &= \left(\frac{\partial P}{\partial T}\right)_\rho \\ dP &= \left(\frac{\partial P}{\partial T}\right)_\rho dT \end{aligned}$$

Plugging this in above yields

$$\begin{aligned} d(\rho_P) &= \left(\frac{\partial \rho_P}{\partial T}\right)_P dT + \left(\frac{\partial \rho_P}{\partial P}\right)_T \left(\frac{\partial P}{\partial T}\right)_\rho dT \\ d(\rho_P) &= \left(\left(\frac{\partial \rho_P}{\partial T}\right)_P + \left(\frac{\partial \rho_P}{\partial P}\right)_T \left(\frac{\partial P}{\partial T}\right)_\rho \right) dT \\ \left(\frac{\partial \rho_P}{\partial T}\right)_\rho &= \left(\frac{\partial \rho_P}{\partial T}\right)_P + \left(\frac{\partial \rho_P}{\partial P}\right)_T \left(\frac{\partial P}{\partial T}\right)_\rho \end{aligned}$$

Now, subbing in the values for the differentials of ρ only in terms of ρ , not its partials,

$$\begin{aligned}\rho_P &= \left(\frac{\partial \rho}{\partial P} \right)_T \\ \left(\frac{\partial \rho_P}{\partial T} \right)_P &= \left(\frac{\partial}{\partial T} \right)_P * \left(\frac{\partial \rho}{\partial P} \right)_T = \left(\frac{\partial^2 \rho}{\partial T \partial P} \right) = \rho_{TP} \\ \left(\frac{\partial \rho_P}{\partial P} \right)_T &= \left(\frac{\partial}{\partial P} \right)_T * \left(\frac{\partial \rho}{\partial P} \right)_T = \left(\frac{\partial^2 \rho}{\partial P^2} \right) = \rho_{PP} \\ \therefore \left(\frac{\partial \rho_P}{\partial T} \right)_\rho &= \rho_{TP} + \rho_{PP} \left(\frac{\partial P}{\partial T} \right)_\rho\end{aligned}$$

Returning to Evaluation of $\left(\frac{\partial^2 P}{\partial T^2} \right)_\rho$ and subbing in the known values from above,
Substituting these results into Equation (9) and using

$$\left(\frac{\partial P}{\partial T} \right)_\rho = -\frac{\rho_T}{\rho_P}$$

gives

$$\left(\frac{\partial^2 P}{\partial T^2} \right)_\rho = \frac{\rho_{TT}\rho_P^2 - 2\rho_T\rho_{TP}\rho_P + \rho_T^2\rho_{PP}}{\rho_P^3} \quad (11)$$

Which is a function of

$$\begin{aligned}\rho_P &= " = fAkima(\Delta P, \Delta \rho, 3, P)" = 0.00864536334 \\ \rho_T &= " = fAkima(\Delta T, \Delta \rho, 3, T)" = -0.03711779101 \\ \rho_{PP} &= " = fAkima(\Delta P, \Delta \rho, 4, P)" = -0.00000221836 \\ \rho_{TT} &= " = fAkima(\Delta T, \Delta \rho, 4, T)" = 0.00140258148\end{aligned}$$

However, ρ_{TP} will need a new data table to be created as it requires a derivative of density with respect to Temperature and then with respect to Pressure. I used the "fakima()" function in order to solve for the differential of $\left(\frac{\partial \rho}{\partial T} \right)_P$.

		d rho / d T					
		Temperature					
Pressure (atm)		0	10	25	50	75	100
		273.15	283.15	298.15	323.15	348.15	373.15
0	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1	0.000	0.000	0.000	0.000	0.000	-0.003	-0.003
3	0.000	0.000	0.000	0.000	0.000	-0.006	-0.006
5	-0.001	-0.001	-0.001	-0.001	-0.001	-0.014	-0.014
10	-0.002	-0.002	-0.001	-0.001	-0.001	-0.029	-0.029
30	-0.007	-0.007	-0.005	-0.004	-0.004	-0.441	-0.441
50	-0.012	-0.011	-0.009	-0.006	-0.006	-0.719	-0.719
100	-0.020	-0.019	-0.017	-0.013	-0.013	-0.647	-0.647
200	-0.039	-0.036	-0.032	-0.025	-0.025	-1.192	-1.194
300	-0.049	-0.046	-0.040	-0.032	-0.032	-1.478	-1.483
400	-0.051	-0.047	-0.043	-0.036	-0.036	-0.670	-0.673
600	-0.051	-0.047	-0.043	-0.038	0.145	0.147	
800	-0.040	-0.037	-0.037	-0.038	-2.637	-2.653	
819.15	-0.041	-0.037	-0.037	-0.038	-2.410	-2.425	
1000	-0.046	-0.040	-0.038	-0.037	3.063	3.083	

Figure 5: Caption

From here I now need to find $\left(\frac{\partial}{\partial P} \right) * \left(\frac{\partial \rho}{\partial T} \right)_P$ using "fakima()" again"

$$\left(\frac{\partial}{\partial P} \right) * \left(\frac{\partial \rho}{\partial T} \right)_P = " = fAkima \left(\Delta P, \Delta \left(\frac{\partial \rho}{\partial T} \right)_P \right) = -0.00001707001$$

From here I was now able to solve for $\left(\frac{\partial^2 P}{\partial T^2}\right)_\rho$ by returning to equation 11

$$\left(\frac{\partial^2 P}{\partial T^2}\right)_\rho = 0.14528082432$$

This gives a new integral for the heat capacity as

$$C_v = C_v^{IG} + \int_\infty^V \left(\frac{\partial^2 P}{\partial T^2}\right)_\rho * T * d\rho$$

C_v^{IG} can be solved for using the CHE Calculator function "CvIG()".

$$C_v^{IG} = " = CvIG(comp1, 10^\circ C)" = 20.87745872$$

Then I converted the integral to be in terms of density.

$$C_v = C_v^{IG} + \int_\infty^V \left(\frac{\partial^2 P}{\partial T^2}\right)_\rho * T * dV$$

$$V = \frac{1}{\rho}$$

$$dV = -\frac{d\rho}{\rho^2}$$

$$C_v = C_v^{IG} + \int_0^\rho \left(\frac{\partial^2 P}{\partial T^2}\right)_\rho * T * dV$$

$$C_v = C_v^{IG} - \int_0^\rho \left(\frac{\partial^2 P}{\partial T^2}\right)_\rho * T * \frac{d\rho}{\rho^2}$$

$$C_v = C_v^{IG} - \left(\frac{\partial^2 P}{\partial T^2}\right)_\rho * T \int_0^\rho \frac{d\rho}{\rho^2}$$

$$C_v = C_v^{IG} - \left(\frac{\partial^2 P}{\partial T^2}\right)_\rho * T * \left[-\frac{1}{\rho}\right]_0^\rho$$

This gives a final expression of

$$C_v = C_v^{IG} + \left(\frac{\partial^2 P}{\partial T^2}\right)_\rho \frac{T}{\rho}$$

And now plugging in

$$T = 283.15K$$

$$C_v^{IG} = 20.87745872 \frac{kJ}{kmol * K}$$

$$\left(\frac{\partial^2 P}{\partial T^2}\right)_\rho = 0.14528082432$$

$$\rho_{819.15atm, 10^\circ C} = 19.656 \frac{kmol}{m^3}$$

$$\therefore C_v = 22.97022555 \frac{kJ}{kmol * K}$$

And now recalling the difference in heat capacities

$$C_p - C_v = 11.7417745 \frac{kJ}{kmol * K}$$

$$C_p = C_v + 11.7417745 \frac{kJ}{kmol * K}$$

$$\therefore C_p = 34.71200006 \frac{kJ}{kmol * K}$$

3.4 H, Molar Enthalpy

Moving on to molar enthalpy. I started by writing out the differential for enthalpy we derived in class.

$$dH = C_P dT + (1 - \beta T) V dP \quad (12)$$

Then I integrated and set the bounds, and assumed constant temperature over the integration,

$$\begin{aligned} \int_{H^{IG}}^H dH &= \int_{P_0}^P (1 - \beta T) V dP \\ H &= H^{IG} + V(1 - \beta T)(P - P_0) \\ H &= H^{IG} + \frac{(1 - \beta T)(P - P_0)}{\rho} \end{aligned}$$

where we can define our variables as

$$H^{IG} = " = HIG(comp1, T = 10^\circ C)" = -437.7625975 \frac{kJ}{kmol}$$

$$T = 10^\circ C = 283.15 K$$

$$\rho_{819.15 atm, 10^\circ C} = 19.656 \frac{kmol}{m^3}$$

$$\beta = 0.001888331 K^{-1}$$

$$P = 819.15 atm = 83,000 kPa$$

$$P_0 = 0 atm = 0 kPa$$

Plugging all known values in gives a value of

$$H = 1,527.067283 \frac{kJ}{kmol} = 1.5271 \frac{kJ}{mol}$$

3.5 S, Molar Entropy

Starting with the entropy differential we solved for in class,

$$dS = C_p \frac{dT}{T} - \beta V dP$$

I then integrated this equation from the ideal gas state to the current system state, and taking it across constant temperature,

$$\begin{aligned} \int_{S^{IG}}^S dS &= - \int_{P_0}^P \beta V dP \\ S - S^{IG} &= -\beta V (P - P_0) \\ S &= S^{IG} - \frac{\beta}{\rho} (P - P_0) \end{aligned}$$

Plugging in our known values below,

$$T = 10^\circ C = 283.15 K$$

$$\rho_{819.15 \text{ atm}, 10^\circ C} = 19.656 \frac{\text{kmol}}{\text{m}^3}$$

$$\beta = 0.001888331 K^{-1}$$

$$P = 819.15 \text{ atm} = 83,000 \text{ kPa}$$

$$P_0 = 1 \text{ atm} = 101.325 \text{ kPa}$$

I used 1 atm instead of 0 as 0 atm gave an undefined answer, the pressure at 1 atm still has a **Compressibility factor** of 1 meaning it still follows ideal gas relations. Then, I solved for S^{IG} by using the excel function "=SIG(comp1, 10°C)"

$$S^{IG} = " = \text{SIG}(\text{comp1}, 10^\circ C, 1 \text{ atm})" = " = \text{SIG}(\text{comp1}, 10^\circ C, 1.01325 \text{ bar})" = -1.506487938$$

Solving for S, entropy now I got

$$S = -9.47031233 \frac{\text{kJ}}{\text{kmol} * K}$$

3.6 μ_{JT} , Joule-Thomson Coefficient

The **Joule-Thomson Coefficient** can be solved for directly with the following equation,

$$\begin{aligned}\mu_{JT} &= \frac{\beta T - 1}{\rho C_p} \\ \beta &= 0.001888331 K^{-1} \\ C_p &= 34.71200006 \frac{kJ}{kmol * K} \\ T &= 283.15 K \\ \rho_{819.15 \text{ atm}, 10^\circ C} &= 19.656 \frac{kmol}{m^3} \\ \mu_{JT} &= \frac{(0.001888331 K^{-1})(283.15 K) - 1}{(19.656 \frac{kmol}{m^3})(34.71200006 \frac{kJ}{kmol * K})} = -0.000681973 \frac{K}{Pa} \\ \mu_{JT} &= -0.000681973 \frac{K}{Pa} = -68.19730677 \frac{K}{bar}\end{aligned}$$

3.7 ν , Speed of Sound

For the speed of sound, I started with the derived definition of

$$\nu^2 = \frac{1}{MW_{N_2}} \gamma \left(\frac{\partial P}{\partial \rho} \right)_T$$

Using the "molw1" function in the CHE Calculator gives the molecular mass of N_2 , and then I used the "fAkima()" function to solve for the partial derivative

$$\begin{aligned}\left(\frac{\partial P}{\partial \rho} \right)_T &= "fAkima(\Delta P, \Delta \rho, \rho_{819.15 \text{ atm}, 10^\circ C}) = 114.7742198 \\ \gamma &= \frac{C_p}{C_v} = \frac{34.71200006}{22.97022555} = 1.51117367\end{aligned}$$

However, this partial was solved across a change in atm, so I have to convert to kPa. Plugging in the above values yields

$$\begin{aligned}\left(\frac{\partial P}{\partial \rho} \right)_T &= 114.7742198 * \frac{101,325 Pa}{1 atm} = 11629497.82 \\ \nu^2 &= \frac{1}{28.013} * 1.51117367 * 11629497.82 = 627358.4016 \frac{m^2}{s^2} \\ \nu &= \sqrt{\nu^2} = \sqrt{627358.4016} = 792.0595947 \frac{m}{s}\end{aligned}$$

4 Question B

The second question of this assignment is about the analysis of a gas venting system of Ethylene. I am asked to solve for the following

1. The initial mass in kg of ethylene in the tank
2. The temperature and pressure at the narrow section of the pipe when the critical mass flow rate which corresponds to the speed of sound (critical condition) is $\dot{m}=5000$ kg/hr.
3. The diameter in mm of the narrow section of the venting pipe
4. The mass in kg of ethylene in the tank when the critical condition is reached
5. When during the venting process the pressure gage reads $P_1=2300$ psig what is the mass in kg of ethylene in the tank

I was also given the following assumptions / hints to make,

1. The venting process is reversible and adiabatic
2. Ethylene behaves as an ideal gas
3. Hint 1 You must start with the unsteady-state mass, energy and entropy balances
4. Hint 2 The kinetic energy of the gas at the narrow section of the pipe cannot be ignored

Within the problem statement, there is a series of data given

$$T_0 = 400K$$

$$P_0 = 200bar$$

4.1 M_0 , Initial Mass of Ethylene

In the problem statement, I am told that the volume of the tank is $20m^3$, and knowing the mass to volume relation

$$m = V * \rho$$

$$m_{Initial} = V_{Initial} * \rho_0$$

Since Volume is given, I have to solve for density under the initial conditions. The problem statement assumes it is an ideal gas giving a value of $Z = 1$ for the Compressibility Factor

$$V = 20m^3$$

$$\rho_0 = " = Density(T_0, P_0, Z_0)" = Density(126.8500^\circ C, 200bar, 1) = 6.014 \frac{kmol}{m^3}$$

Then I multiplied out to get,

$$m_{Initial} = m_0 = (20m^3) * (6.014 \frac{kmol}{m^3}) = 120.2724kmol$$

Then i used the CHE Calculator function "molw1 = 28.0540 $\frac{kg}{kmol}$ " to get the molar mass of Ethylene.

$$m(kg) = n(kmol) * M_{W,Ethylene} = 120.2724kmol * 28.0540 \frac{kg}{kmol} = 3,374.1206kg$$

4.2 System Mass Balance

Within the question, the mass flow rate of the narrowed section is given as 5,000 $\frac{kg}{hr}$ or 1.3889 $\frac{kg}{s}$
I started with the unsteady-state equation for mass balance

$$Accumulation = in - out$$

$$\frac{dm_{sys}}{dt} = \dot{M}_{in} - \dot{M}_{out}$$

since there is no influent mass into the system the equation becomes

$$\frac{dm_{sys}}{dt} = \frac{dm}{dt} = -\dot{m}_s$$

$$\frac{dm}{dt} = -\dot{m}_s \quad (13)$$

4.3 T_C and P_C , Temperature and Pressure under critical conditions

Locally at the point of the nozzle it acts steady and starting with the equation for stagnant flow from a very large source at the throttle point

$$\begin{aligned} H_0 &= H_s + \frac{v_s^2}{2} \\ C_p T_0 &= C_p T_s + \frac{v_s^2}{2} \end{aligned} \quad (14)$$

taking this relation and applying it the the equation above returns the following formula

$$v^2 = \frac{1}{M_w} \gamma \left(\frac{\partial P}{\partial \rho} \right)_T \quad (15)$$

since this is assumed to be ideal, then the ideal gas law applies

$$\begin{aligned} PV &= RT_s \\ P &= \rho RT_s \\ \left(\frac{\partial P}{\partial \rho} \right)_T &= RT_s \\ \nu^2 &= \frac{\gamma RT_s}{M_w} \end{aligned} \quad (16)$$

where

$$\begin{aligned} R_* &= \frac{R}{M_w} \\ R &= 8.314 \frac{kJ}{kmol * K} \\ \gamma &= \frac{C_p}{C_v} = \frac{"C_p IG(comp1, T)"}{"C_v IG(comp1, T)"} = \frac{54.2060}{45.8916} = 1.1812 \\ M_w &= "molw1" = 28.0540 \frac{kg}{kmol} \\ \therefore \nu^2 &= \gamma R_* T_s \end{aligned} \quad (17)$$

and plugging into equation 12 yields

$$\begin{aligned} C_p T_0 &= C_p T_s + \frac{\gamma R_* T_s}{2} \\ T_0 &= T_s + \frac{\gamma R_* T_s}{2 * C_p} \end{aligned}$$

Plugging in the known ideal gas heat capacity relation

$$C_p = \frac{R_* \gamma}{\gamma - 1}$$

Plugging in and simplifying receives

$$\begin{aligned} T_0 &= T_s + \frac{\gamma R_* T_s}{2 * \frac{\gamma R_*}{\gamma - 1}} \\ T_0 &= T_s + \frac{\gamma R_* T_s (\gamma - 1)}{2 \gamma R_*} \\ T_0 &= T_s + \frac{T_s (\gamma - 1)}{2} \\ T_0 &= T_s \left(1 + \frac{1}{2} (\gamma - 1) \right) \\ T_s &= \frac{2T}{\gamma + 1} = \frac{2 * 400k}{1.1812 + 1} = 366.7746k \end{aligned}$$

The temperature of the outlet stream from the tank is 366.7746 K.

4.4 System Entropy Balance

Starting with the unsteady balance format,

$$\frac{dS_{sys}}{dt} = -\Delta mS + \sum \frac{\dot{Q}_i}{T_i} + S_{gen}$$

$$\frac{dS_{sys}}{dt} = \dot{m}_0 * S_0 - \dot{m}_1 * S_1 + \sum \frac{\dot{Q}_i}{T_i} + S_{gen}$$

This is the same case as the mass balance where there is no influent. In addition, I assumed reversible and adiabatic so the equation becomes

$$\frac{dS_{sys}}{dt} = -\dot{m}_1 s_1 = \frac{d(ms)}{dt}$$

expanding the right side gives

$$\frac{d(ms)}{dt} = m \frac{ds}{dt} + s \frac{dm}{dt}$$

and from the mass balance (equation 11), we know that

$$\frac{dm_{sys}}{dt} = \frac{dm_s}{dt} = -\dot{m}_1$$

$$\therefore \frac{dS_{sys}}{dt} = \frac{d(ms)}{dt} = m \frac{ds}{dt} + s \frac{dm}{dt} = -\dot{m}_1 s_1$$

$$m \frac{ds}{dt} + s \frac{dm}{dt} = -\dot{m}_1 s_1$$

$$m \frac{ds}{dt} + s * -\dot{m}_1 = -\dot{m}_1 s_1$$

$$m \frac{ds}{dt} = 0$$

$$\frac{ds}{dt} = 0$$

The change in entropy over time through the system is equal to zero and therefore the entropy inside the tank is equal to the entropy exiting.

For choked, isentropic flow from a large reservoir,

$$\frac{P^s}{P_0} = \left(\frac{2}{\gamma + 1} \right)^{\frac{\gamma}{\gamma - 1}}$$

Thus,

$$P_s = P_0 * \left(\frac{2}{\gamma + 1} \right)^{\frac{\gamma}{\gamma - 1}}$$

Knowing that $\gamma = 1.1812$, the pressure ratio is

$$P^s = 200 \text{ bar} \times 0.5682 = 113.6321 \text{ bar.}$$

This gave a pressure, P_s , of 113.6321 bar for the effluent stream.

4.5 Diameter and Area of Narrowed section

The area can be found from the equation for mass flow

$$\dot{m}_s = A_{narrow} * v * \rho_s$$

at the critical conditions, I know that the mass flow is $5000 \frac{kg}{hr}$ and the velocity is the speed of sound under these conditions.

From equation 15, the velocity equation was established as

$$\begin{aligned} \nu^2 &= \gamma R_* T_s \\ \nu &= \sqrt{\gamma R_* T_s} = \sqrt{\gamma \frac{R}{M_W} T_s} = \sqrt{1.1812 * \left(0.2964 \frac{kJ}{kg * K} * 1000 \frac{J}{kJ} \right) * 366.7746 K} = 358.3246 m/s \\ \rho_s &= \rho = \text{Density}(T_s, P_s, Z = 1) = 3.7262 \frac{kmol}{m^3} \\ \dot{m} &= 5000 \frac{kg}{hr} * \frac{1hr}{3600sec} = 1.3889 \frac{kg}{s} \end{aligned}$$

Returning to the mass flow equation,

$$\therefore 1.3889 = A * (358.3246 \frac{m}{s}) * (3.7262 \frac{kmol}{m^3}) * (M_w = 28.0540 \frac{kg}{kmol})$$

$$\therefore A = 0.000037 m^2 = 37.0790 mm^2$$

$$A = \frac{\pi}{4} d^2 \quad \therefore d = 6.8710 mm$$

4.6 Mass remaining in the tank under critical conditions

Starting with ideal gas isentropic relation

$$PV^\gamma = \text{constant}$$

$$P_0 V_0^\gamma = PV^\gamma$$

$$P_0 \left(\frac{V_{\text{tank}}}{m_0} \right)^\gamma = P \left(\frac{V_{\text{tank}}}{m} \right)^\gamma$$

since the volume of the tank is held constant as its rigid,

$$\frac{P_0}{(m_0)^\gamma} = \frac{P}{(m)^\gamma}$$

$$\frac{P}{P_0} = \left(\frac{m}{m_0} \right)^\gamma$$

$$\frac{m}{m_0} = \left(\frac{P_0}{P} \right)^{\frac{1}{\gamma}}$$

$$m = m_0 \left(\frac{P_0}{P} \right)^{\frac{1}{\gamma}} \quad (18)$$

Knowing the initial mass and pressure,

$$m = 3,374.1206 \left(\frac{P}{200 \text{ bar}} \right)^{\frac{1}{\gamma}}$$

Under the throttling conditions, The pressure is equal to the pressure in the tank which was solved for as $P = 113.6321$ bar. Plugging these values in gave,

$$m = 2,090.7038 \text{ kg @ } P = 113.6321 \text{ bar}$$

4.7 Mass when the narrow gage reads 2300 psig

Equation 16 may be used to calculate thus mass also. In this case $P = 2300$ psig. Starting with unit conversions.

$$2300psig = \text{gage pressure}$$

I need to convert to absolute pressure

$$P_{gage} = P_{absolute} - P_{atmosphere}$$

$$P_{absolute} = P_{gage} + P_{atmosphere}$$

$$P_{absolute} = 2,300psig + (P_{atm} = 14.6959psig) = 2,314.6959psia$$

$$2,314.6959psia = 159.5921bar$$

Recalling equation 16,

$$m = m_0 \left(\frac{P}{P_0} \right)^{\frac{1}{\gamma}}$$

$$m_p = (3,374.1206kg) \left(\frac{159.5921bar}{200bar} \right)^{\frac{1}{1.1812}} = 2,787.2552kg \text{ remaining}$$

5 Conclusion

The goal of this report was to A.) Perform an Analysis on the Thermal Properties of nitrogen and B.) Analyze Ideal Gas Venting of Ethylene. Through this report, I solved for the density of nitrogen as a function of both pressure and temperature. I then used this data to solve for the Molar Enthalpy, Molar Entropy, both constant volume and pressure Heat Capacities, Thermal Expansion Coefficient, Pressure Expansion Coefficient, the Speed of Sound, and the Joule-Thomson Coefficient. Through ideal gas and isentropic relations, the venting of ethylene was analyzed and solved for the mass and pressure as a function of its thermal properties.

6 References

This report used sources from the following

1. ChatGPT was used to interpret and help code the document into LaTeX code. In addition, it was used to provide guidance and direction on this assignment and provide better structure and readability to the report.
2. NASA website data on critical flow conditions
3. CHE 301 Assignment #03 handout (Compressibility factor table for N_2)